

HackRx 6.0

Ideate • Co-create • Impact



EduSenseAI

An AIML and LLM-based education analysis platform

HackRx 6.0

Ideate • Co-create • Impact



Team OG

- **Namboori Harshith Varma** | Graduating year: 2028 | Chennai Institute Of Technology
- **Mithun R** | Graduating year: 2028 | Chennai Institute Of Technology
- **Kaushik Krushnan A** | Graduating year: 2028 | Chennai Institute Of Technology
- **Gadepalli Saketh** | Graduating year: 2028 | Chennai Institute Of Technology

HackRx 6.0

Ideate • Co-create • Impact



Tell us a bit about yourself

We, the team OG, from Chennai Institute of Technology, are a group of enthusiastic students who love learning and building new things. This is our first hackathon, and we are excited to challenge ourselves, try out fresh ideas, and work together as a team. Each of us brings different skills and perspectives, and we're looking forward to combining them to create something useful and creative. More than anything, we want to learn from the experience and have fun while doing it.

HackRx 6.0

Ideate • Co-create • Impact

Problem statement

Problem

- Exam reports are mark-centric and do not guide improvement.
- Students have no clarity on:
 - **What concepts they struggle with**
 - **Why they lose their marks**
 - **How to prepare smarter next time**

HackRx 6.0

Ideate • Co-create • Impact

Problem statement

Solution For The Problem:

EduSenseAI leverages **LLM**, **voice analysis**, and **interaction patterns** to:

- Analyze the **written** and **MCQ** exam answers.
- Detect the **conceptual gaps** based on analysis of the topper's and average student's **answer sheet**.
- Provide clear feedback regarding a student's **strengths** and **weaknesses** and provide a personalized modules for improvements along with external links based on the weak topics for improvements.

HackRx 6.0

Ideate • Co-create • Impact

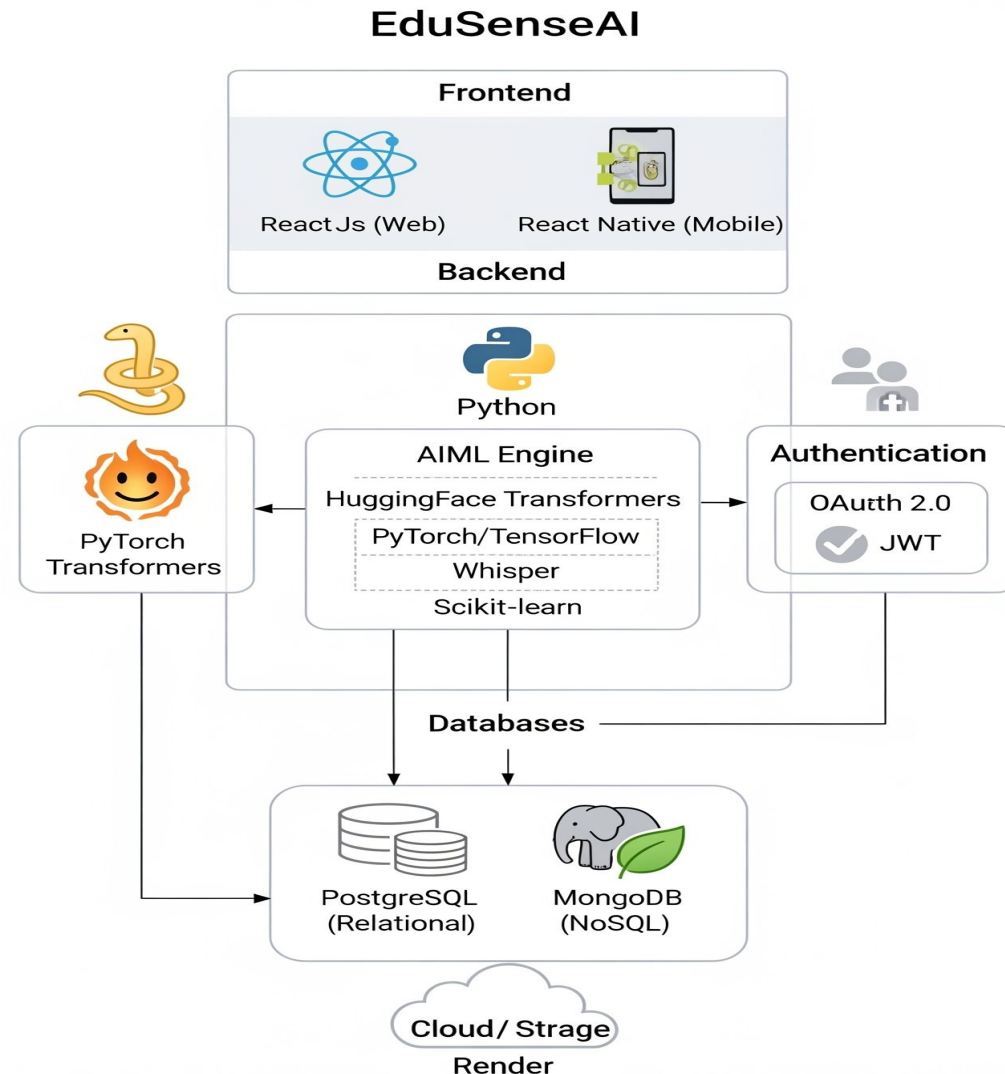
Tech Stack

- Frontend: **ReactJs** (for web), **react Naive** (for mobile app)
- Backend: AIML engine with **Python**
 - **HuggingFace Transformers**: LLM Based answer evaluation
 - **PyTorch/TensorFlow**: Custom classification and analytics
 - **Whisper**: voice transcription
 - **Scikit-learn**: Student behaviour modeling
- Authentication: OAuth 2.0, +JWT
- Database: **PostgreSQL**(Structured data), **MongoDB**(logs and model feedback)
- Cloud and storage: **Render**

HackRx 6.0

Ideate • Co-create • Impact

Detailed description of the solution



HackRx 6.0

Ideate • Co-create • Impact



Detailed description of the solution

How the application works:

Step 1: The student's answer sheets are uploaded as pdf into the application by the administration team of the college or the students themselves based on the option chosen by the college.

Step 2: After the answer sheets are uploaded, the LLM model detects the mistakes done by the student in the answer sheet. Also, it analyses the marks of the top marks secured student of that particular class and also of average student. The student and teacher of that class will be able to see his performance.

Step 3: Once all the analysis are done by the LLM model, the AIML model in the application creates the roadmaps for the students in places where he/she is lacking along with few external links from various education websites and youtube for explanation in that topic.

HackRx 6.0

Ideate • Co-create • Impact



Data Flow Diagram



EduSenseAI

Reports (Visible to Educators & Students)



External

EduSenseAI

EduSenseAI System



Performance Analysis (Visible to Teachers)



External Entities



Educators



Students

Teachers

HackRx 6.0

Ideate • Co-create • Impact



So, how is your solution different?

First-of-its-kind student performance system that uses:

- **LLM-based evaluation** for written answers, where we convert the written answers into PDF format and upload it into the model
- **Voice analysis** for oral exams to detect emotional signals for **stress management**
- **Personalized** practice generation
- Scalable, role-based access for institutions
- **Easy integration** into existing school ecosystems



HackRx 6.0

Ideate • Co-create • Impact



FINSERV

Future possible enhancements

- Add AI chatbots for solving **teacher-student** query resolution
- **National Curriculum Alignment** (CBSE, NEP, UGC)
- AI-based **career or course recommendation** engine for future.

HackRx 6.0

Ideate • Co-create • Impact



Risks/Challenges/Dependencies

- **OCR accuracy** on handwritten answers may vary
- Privacy concerns with **student voice data**
- Ensuring **AI fairness** across different student demographics
- Institution's **willingness** to adopt AI-based analytics
- Large-scale LLM **inference costs** (solved via local models)

HackRx 6.0

Ideate • Co-create • Impact



Acceptance Criteria Coverage

- Supports **subjective, objective, and voice-based exams**
- Provides individualized **performance analysis**
- Offers **teacher-level insights**
- Scalable for **multiple institutions**
- Includes AI-generated **feedback and practice**
- **Role-based** access control and data security

HackRx 6.0

Ideate • Co-create • Impact



Conclusion:

- EduSenseAI is designed with **real classroom needs** in mind.
- It **reduces manual effort** for teachers and boosts student outcomes through targeted, explainable insights.
- We believe it can transform how **assessments are understood, acted upon, and used to drive learning success**.